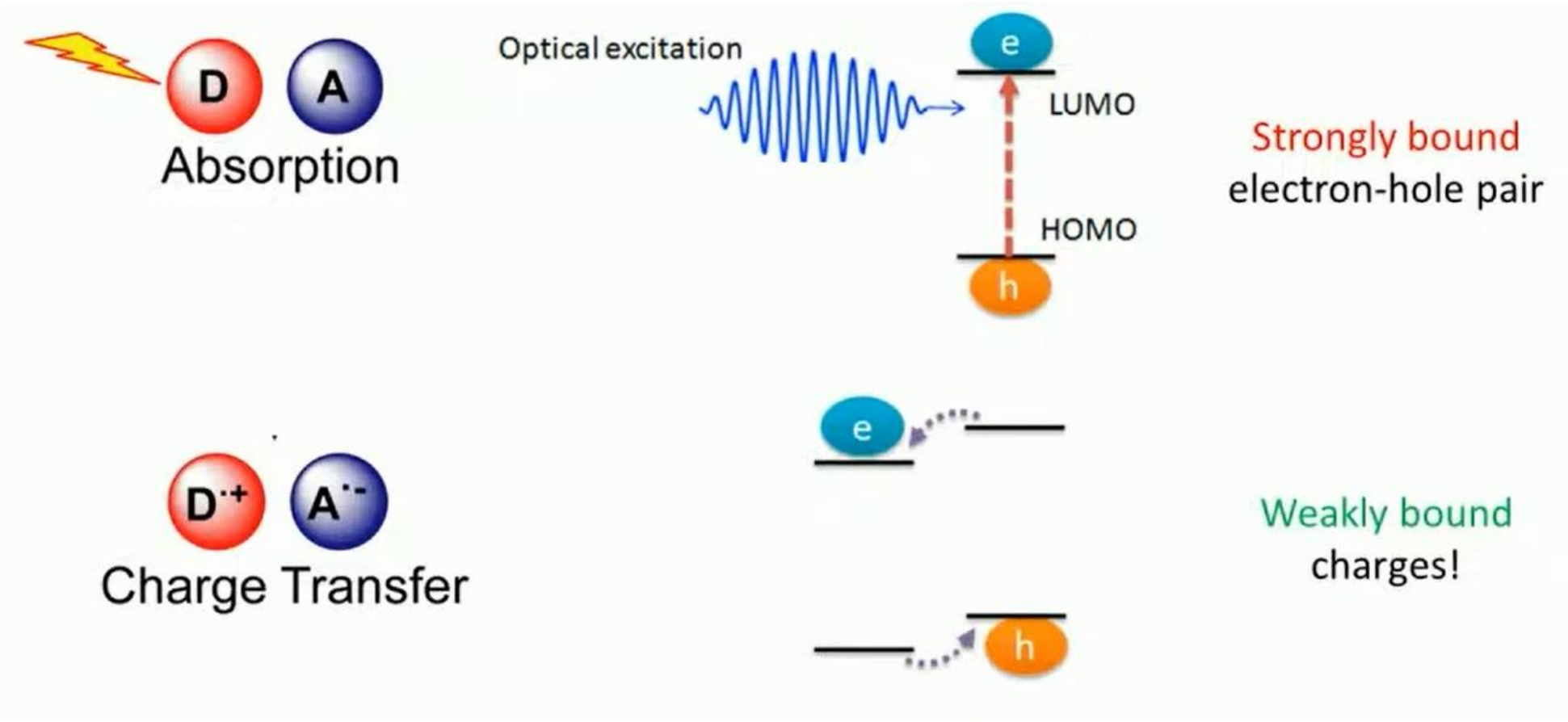
Electron Transfer: Molecular Perspective



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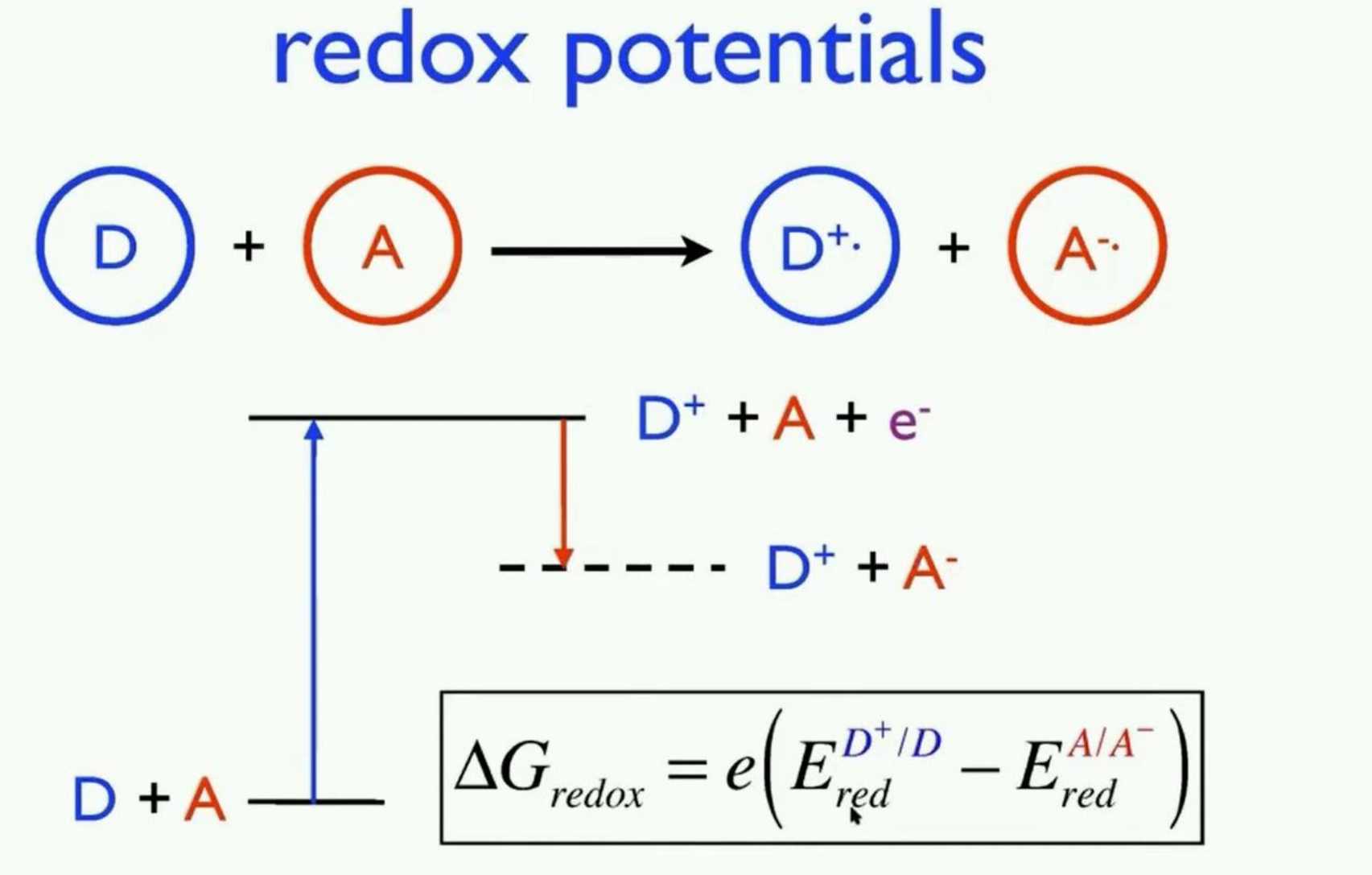
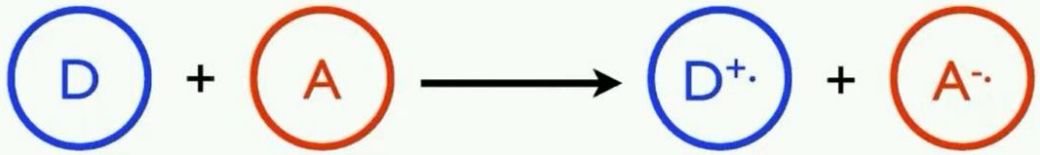


$$\Delta G_{redox} = e\left(E_{red}^{D^+/D} - E_{red}^{A/A^-}\right)$$

Electron Transfer: Molecular Perspective

Solvent correction of redox potentials



$$\Delta G_{redox} = e\left(E_{red}^{D^+/D} - E_{red}^{A/A^-}\right)$$

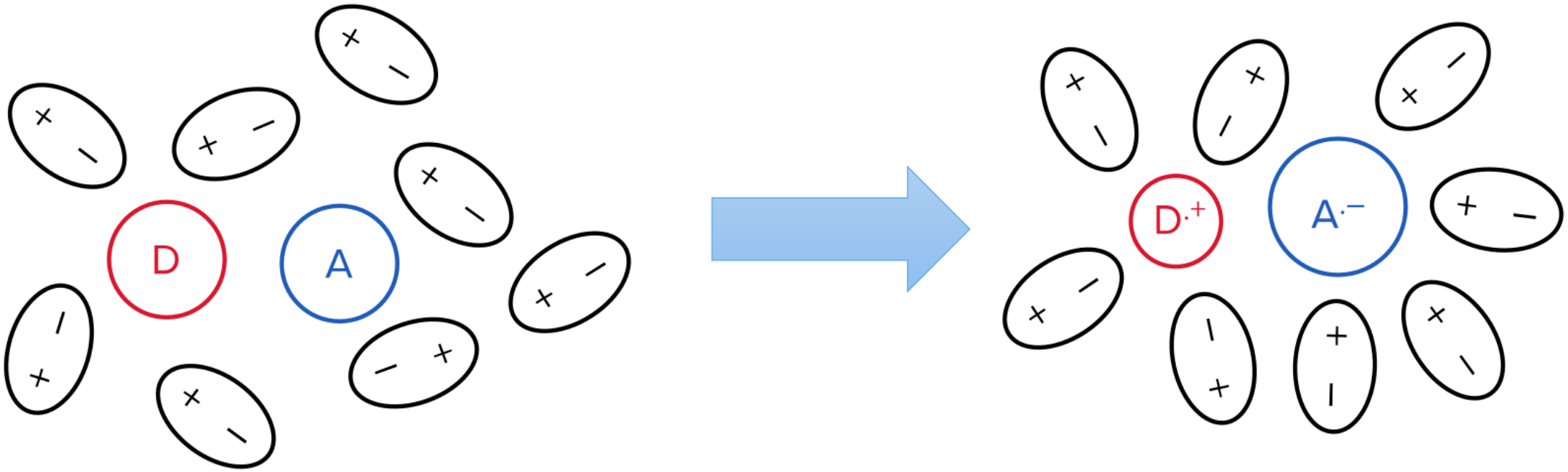
- Usually measured in polar solvent, but ion pair can exist in less polar solvent! Correction:

$$\Delta G_{redox} = e(E_{red}^{D^+/D} - E_{red}^{A/A^-}) - \frac{e^2}{r_{ion}}\left[\left(1 - \frac{1}{\varepsilon_{solvent}}\right) - \left(1 - \frac{1}{\varepsilon_{polar}}\right)\right]$$

$$\Delta G_{redox} = e(E_{red}^{D^+/D} - E_{red}^{A/A^-}) - \frac{e^2}{r_{ion}}\left(\frac{1}{\varepsilon_{polar}} - \frac{1}{\varepsilon_{solvent}}\right)$$

Electron Transfer: Molecular Perspective

On the molecular level, electron transfer involves (1) ***inner-sphere*** reorganization of bonds and atoms in D and A and (2) ***outer-sphere*** reorganization of solvent molecules.



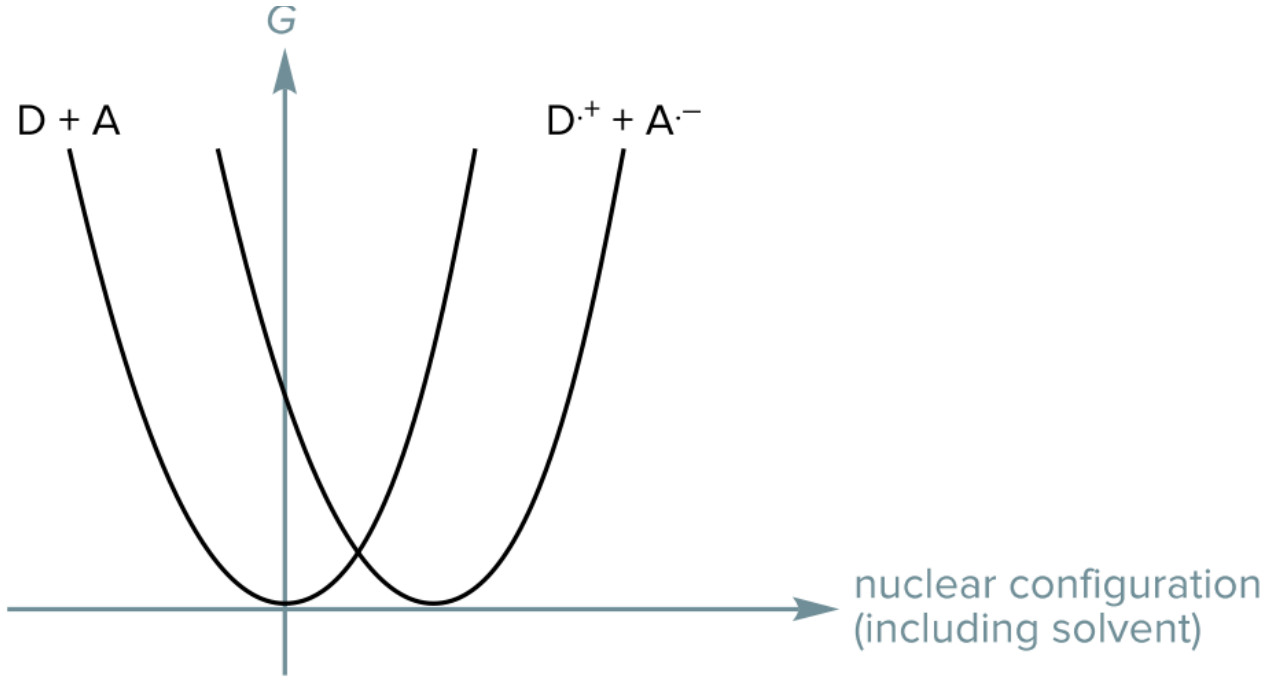
Do inner- and outer-sphere motions happen in concert or does inner-sphere electron transfer happen first?

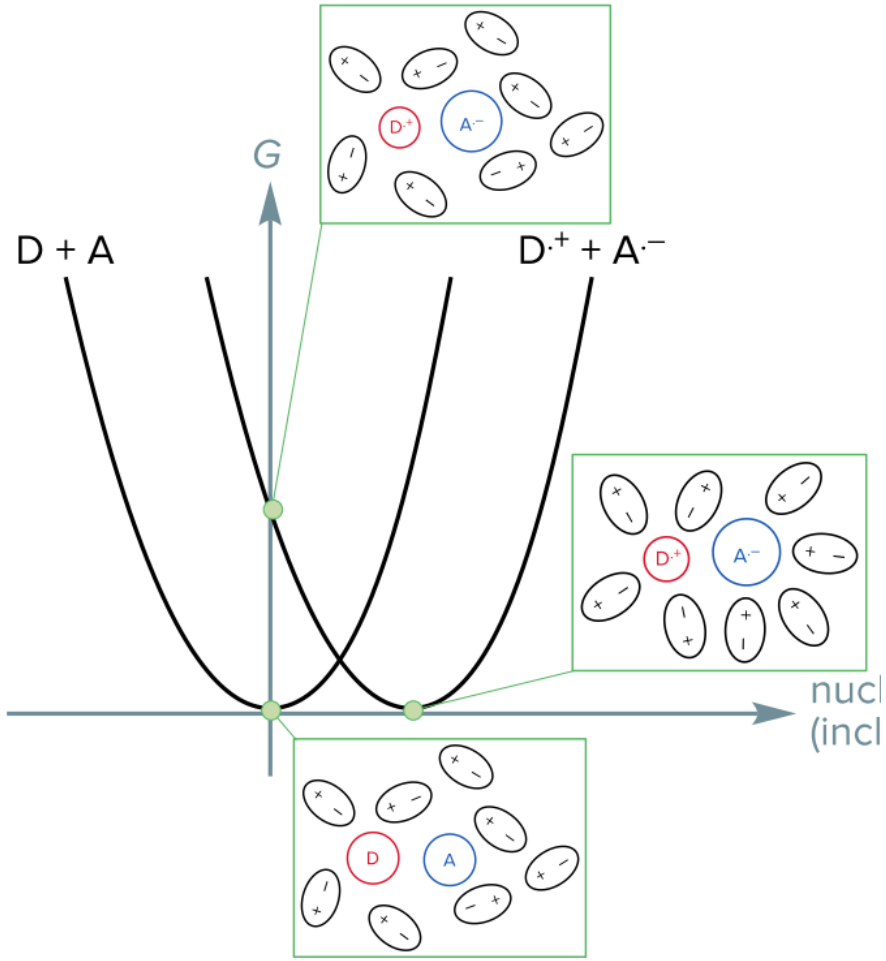
Electron Transfer: Libby Theory

The energy released as solvent adjusts to D^+ and A^- is called ***reorganization energy (λ)***. Libby posited that λ must be fully invested before the product PES is reached.

Self-exchange reaction ($\Delta G^o = 0$)

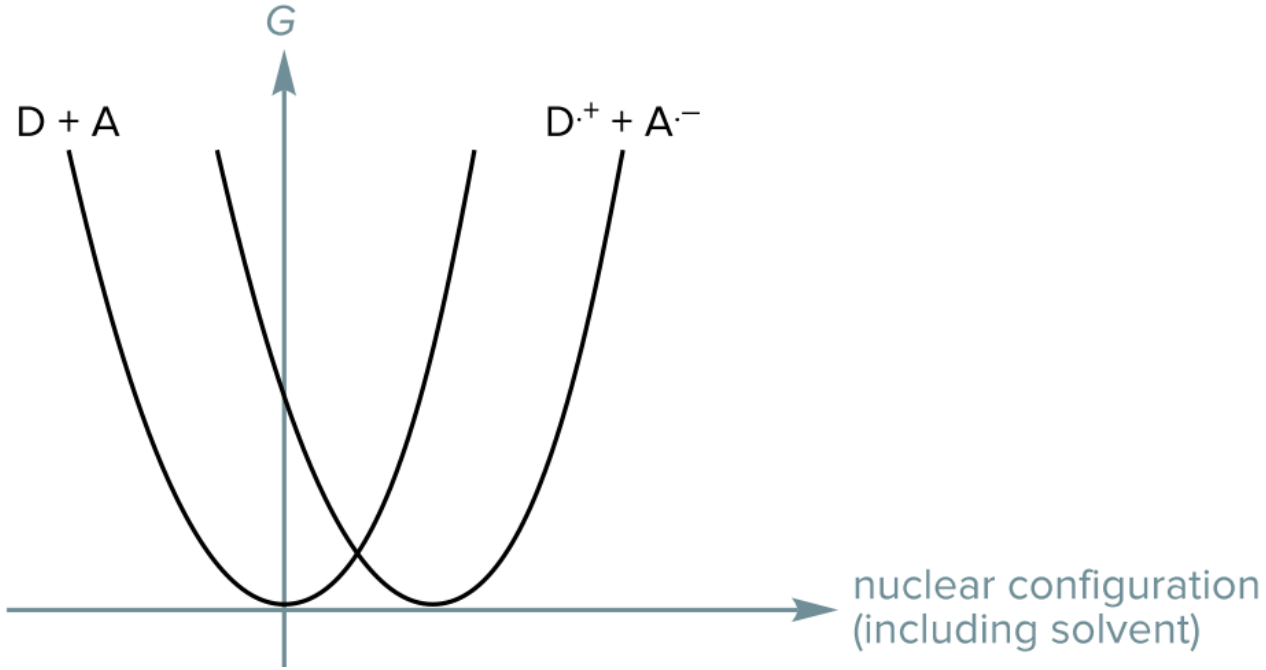
Hypothesis: $\Delta G^\ddagger = \lambda$





Electron Transfer: Marcus Theory

If solvent molecules move in concert with the transferred electron, the activation energy is lower than that predicted by the Libby model. The resulting picture is the foundation of Marcus theory.



Self-exchange reaction ($\Delta G^o = 0$)

Hypothesis: $\Delta G^{\ddagger} = \lambda/4$

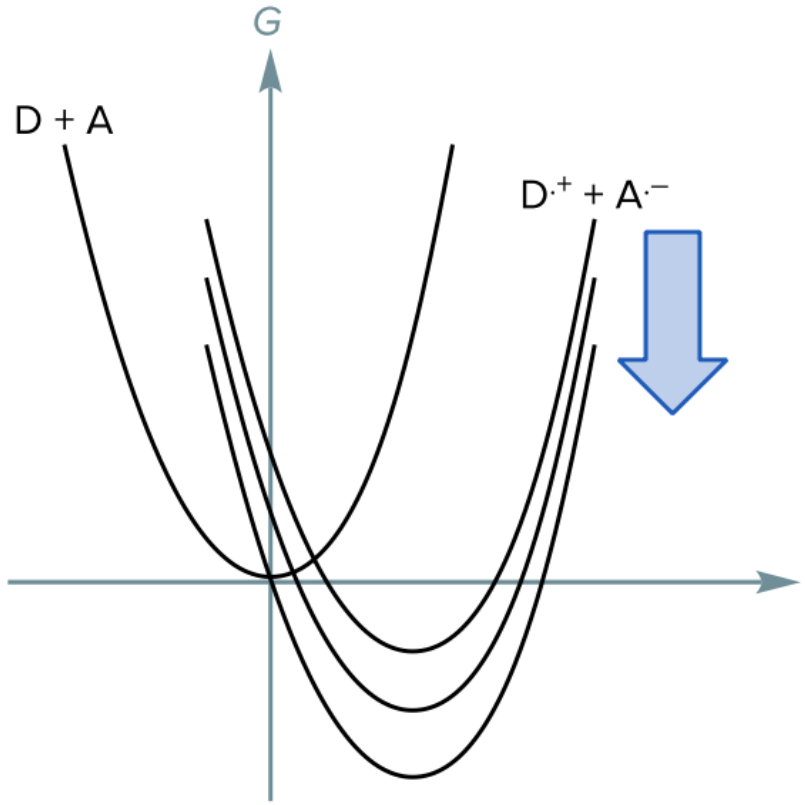
What happens if $\Delta G^o \neq 0$?

1956-Proposed, 1992 Nobel

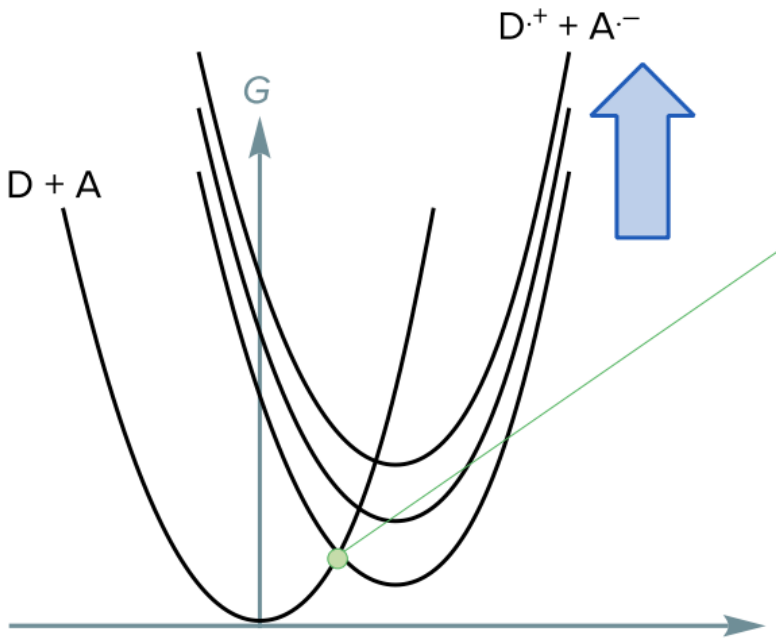
Electron Transfer: Marcus Theory

Endergonic ET reactions behave as expected: $\Delta G^{\ddagger}$ increases as ΔG° increases. Until $\Delta G^{\circ} = -\lambda$, exergonic ET reactions also behave as expected.

Exergonic electron transfer



Endergonic electron transfer



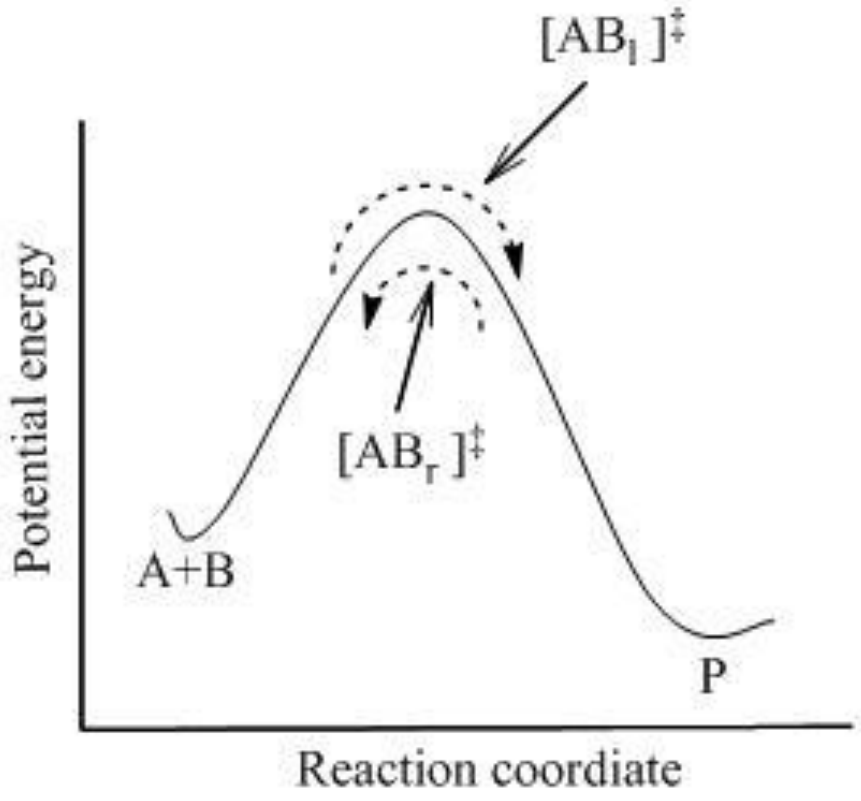
$$G_{\mathrm{DA}} = x^2$$

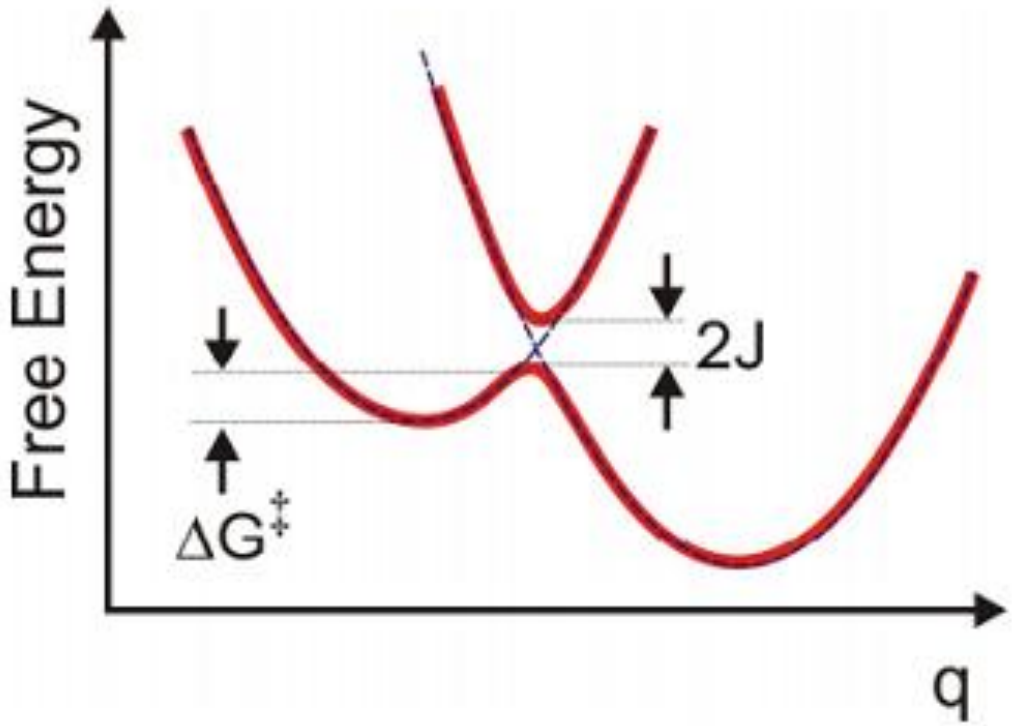
$$G_{\mathrm{D^+A^-}} = (x - \sqrt{\lambda})^2 + \Delta G^{\circ}$$

Where the G values are equal...

$$\Delta G^{\ddagger} = \frac{(\lambda + \Delta G^{\circ})^2}{4\lambda}$$

Electron Transfer: Marcus Theory (Quantum + Arrehenius)

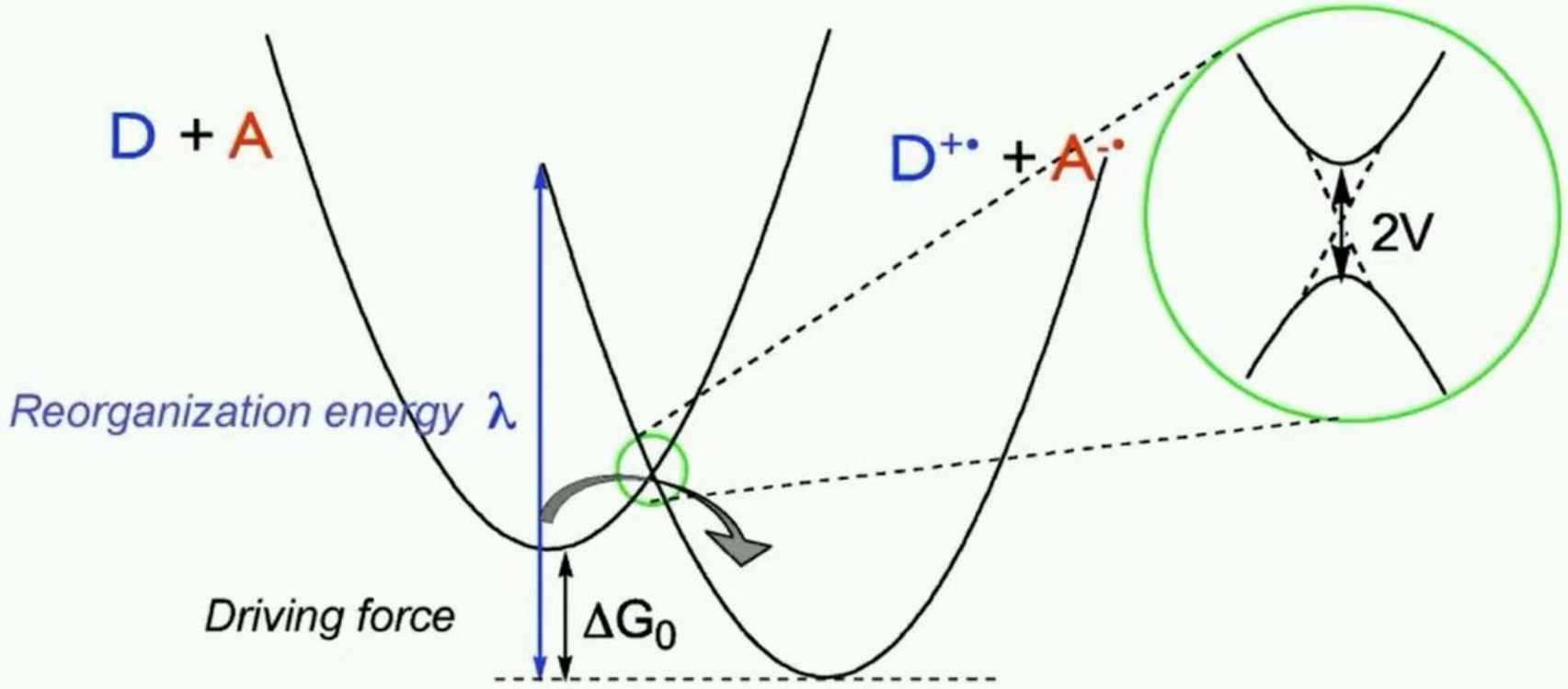




$$k = Ae^{\frac{-E_a}{RT}} = A\exp\left(\frac{-E_a}{RT}\right)$$

$$k_{ET} = A\exp\left(-\frac{\Delta G^{\ddagger}}{k_B T}\right)$$

Electron Transfer: Marcus Theory (Quantum + Arrehenius)



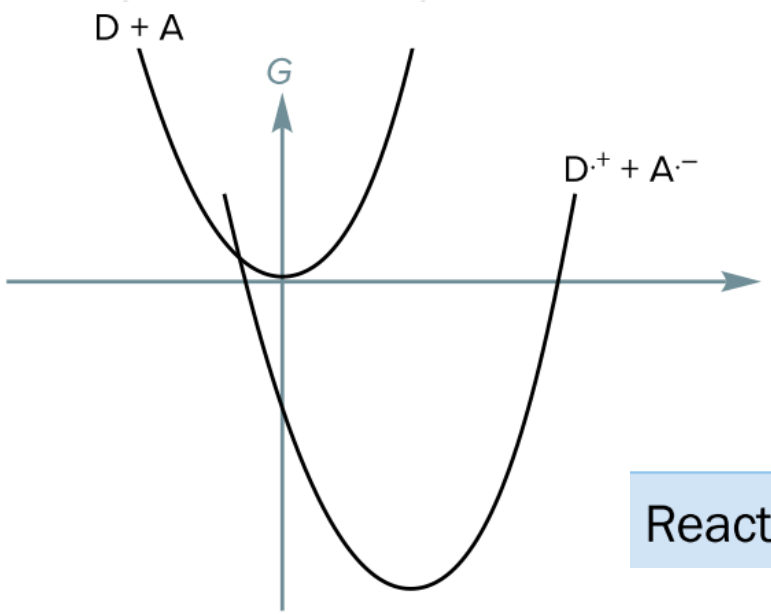
$$k_{et} = \frac{2\pi^{3/2}}{h\sqrt{\lambda k_B T}} V^2 e^{-(\Delta G_0+\lambda)^2/4\lambda k_B T}$$

Electron Transfer: Marcus Theory (Very Exergonic System)

Something very strange happens when $\Delta G^o < -\lambda$. The activation energy of electron transfer starts to increase as the reaction becomes more exergonic!

$$\Delta G^{\ddagger} = \frac{(\lambda + \Delta G^{\circ})^2}{4\lambda}$$

Very exergonic electron transfer

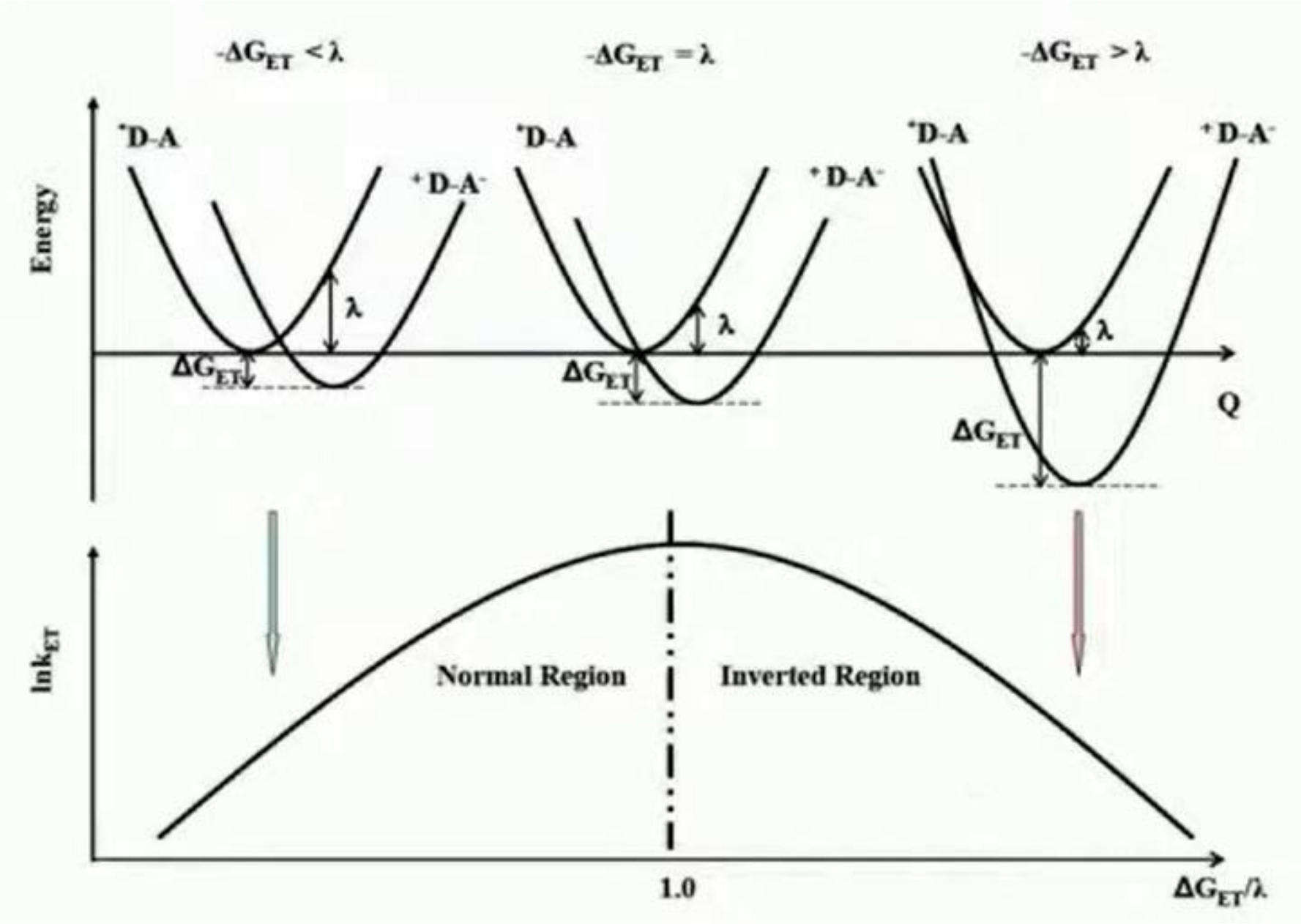


Maximal rate of electron transfer when $\Delta G^o = -\lambda$

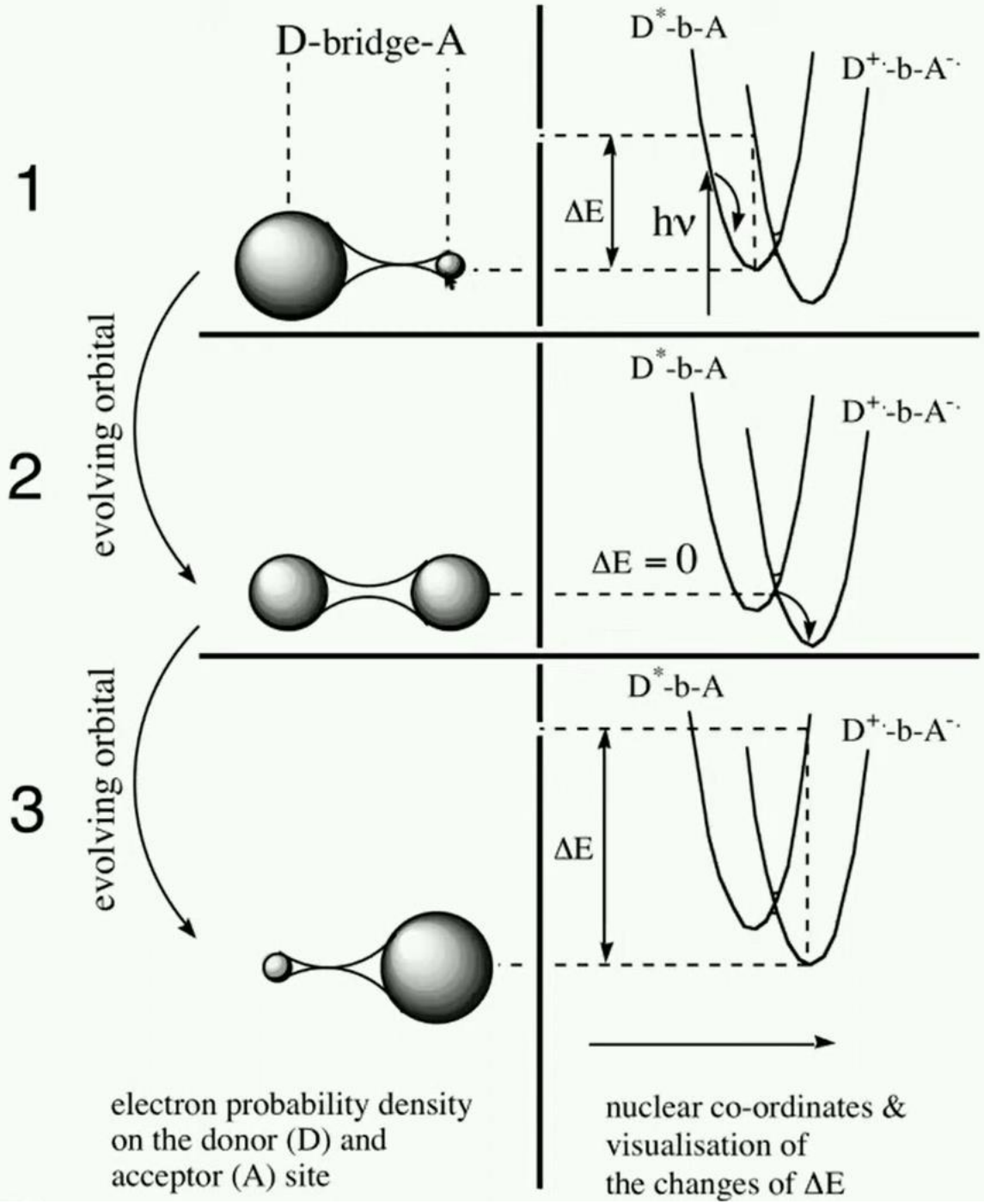
Good news for photoinduced electron transfer...

Reactions for which $\Delta G^o < -\lambda$ are said to be in the ***Marcus inverted region***.

Regions of Electron Transfer: Marcus Theory



Electron Transfer: Marcus Conclusion



1 After excitation the electron is still mainly localised on D, but there is already a small probability on A.

2 At the crossing point (the barrier) there is an equal chance of finding the electron on both sides; $\Delta E = 0$, the electron is formally transferred.

3 After relaxation into the well of the CT state the probability to find the electron on the A side is highest. ΔE has again increased sharply.